SIMMAK PPKP

Diagram Pack — ERD, DFD, BPMN, Architecture

PPKP Provinsi DKI Jakarta

Sistem Informasi Manajemen Mutu, Audit Internal dan Akreditasi

Tanggal: 25 Juni 2026 | Versi Blueprint: 1.0

Laravel 12 | PostgreSQL 17 | Redis | Docker | LDAP AD DKI

SIMMAK PPKP — Diagram Pack

Kumpulan diagram arsitektur dan proses bisnis.

ARCHITECTURE

[Diagram Mermaid — lihat file diagrams/*.mmd untuk versi visual]

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%% ═══════════════════════════════════════════════════════════════════════ %% SIMMAK PPKP — Solution Architecture Diagram %% Sistem Informasi Manajemen Mutu, Audit Internal dan Akreditasi %% Organization: PPKP Provinsi DKI Jakarta | Laravel 12 | PostgreSQL 17 %% Version: 1.0.0 | Deployment: Docker Compose on Ubuntu Server %% ═══════════════════════════════════════════════════════════════════════ %%{init: { 'theme': 'base', 'themeVariables': { 'primaryColor': '#EBF8FF', 'primaryTextColor': '#1A365D', 'primaryBorderColor': '#3182CE', 'lineColor': '#4A5568', 'secondaryColor': '#F0FFF4', 'tertiaryColor': '#FAF5FF', 'fontFamily': 'Segoe UI, Inter, sans-serif', 'fontSize': '13px' }, 'flowchart': { 'htmlLabels': true, 'curve': 'basis', 'padding': 16 } }}%% flowchart TB %% ═══════════════════════════════════════════════════════════════ %% LAYER 1: CLIENT TIER %% ═══════════════════════════════════════════════════════════════ subgraph CLIENT["■■ CLIENT LAYER — Presentation Tier"] direction TB subgraph WEB["Web Application"] BROWSER["Web Browser<br/>(Chrome / Edge / Firefox)"] SPA["Blade + Livewire / Inertia.js<br/>Responsive UI — Desktop & Tablet"] PWA["Progressive Web App<br/>(Offline-capable dashboards)"] end subgraph DASHBOARDS["Role-Based Dashboards"] EXEC_DASH["Executive Dashboard<br/>Readiness · CAPA · Risk Heatmap · Rankings"] CLINIC_DASH["Clinic Dashboard<br/>Audit · CAPA · Risiko · Akreditasi · Mutu"] ACCRED_DASH["Accreditation Dashboard<br/>Self Assessment · Gap · Mock Survey"] end subgraph CLIENT_MODULES["Application Modules"] MOD_DMS["DMS Module"] MOD_AUDIT["Internal Audit Module"] MOD_CAPA["CAPA Module"] MOD_RISK["Risk Management Module"] MOD_ACCRED["Accreditation Module"] MOD_QUALITY["Quality Management Module"] MOD_ADMIN["System Administration"] end BROWSER --> SPA SPA --> PWA SPA --> DASHBOARDS SPA --> CLIENT_MODULES end %% ═══════════════════════════════════════════════════════════════ %% LAYER 2: EDGE / REVERSE PROXY %% ═══════════════════════════════════════════════════════════════ subgraph NGINX_LAYER["■ NGINX REVERSE PROXY — Edge Tier"] direction TB NGINX["Nginx 1.27+<br/>Reverse Proxy & Load Balancer"] SSL["SSL/TLS Termination<br/>Let's Encrypt / Internal CA"] GZIP["Gzip / Brotli Compression"] RATE["Rate Limiting & DDoS Protection"] STATIC["Static Asset Serving<br/>CSS · JS · Fonts · Images"] UPSTREAM["Upstream Pool<br/>PHP-FPM Socket / TCP"] NGINX --> SSL NGINX --> GZIP NGINX --> RATE NGINX --> STATIC NGINX --> UPSTREAM end %% ═══════════════════════════════════════════════════════════════ %% LAYER 3: APPLICATION TIER — LARAVEL 12 %% ═══════════════════════════════════════════════════════════════ subgraph LARAVEL["■ LARAVEL APPLICATION LAYER — Business Logic Tier"] direction TB subgraph ENTRY["Request Pipeline"] PHP_FPM["PHP 8.4-FPM<br/>Process Pool"] KERNEL["HTTP Kernel<br/>Middleware Stack"] ROUTES["Route Layer<br/>Web · API v1 · Health"] end subgraph MIDDLEWARE["Cross-Cutting Middleware"] MW_AUTH["Authenticate<br/>Local + LDAP Session"] MW_RBAC["RBAC Authorization<br/>Spatie Permission / Custom"] MW_SCOPE["Org/Clinic Scope Filter<br/>organization_id · clinic_id"] MW_AUDIT["Audit Trail Middleware"] MW_THROTTLE["Throttle & CSRF Protection"] end subgraph APP_CORE["Application Core"] CONTROLLERS["Controllers<br/>Web · API · Admin"] SERVICES["Domain Services<br/>DMS · Audit · CAPA · Risk · Accreditation · Quality"] REPOS["Repository Layer<br/>Eloquent ORM Abstraction"] EVENTS["Events & Listeners<br/>Domain Event Bus"] POLICIES["Policies & Gates<br/>Fine-grained Authorization"] end subgraph WORKFLOW_NOTIF["Workflow & Notification Engine"] WF_ENGINE["Workflow Engine<br/>Dynamic Step Configuration"] WF_STATE["State Machine<br/>Draft → Review → Approve → Publish"] NOTIF_SVC["Notification Service<br/>In-App · Email · Queue"] SCHEDULER["Laravel Scheduler<br/>Expiry Alerts · SLA Escalation"] end subgraph BACKGROUND["Background Processing"] QUEUE_WORKER["Queue Workers<br/>Horizon / Supervisor"] JOBS["Async Jobs<br/>LDAP Sync · Report Gen · Email · Export"] CACHE_SVC["Cache Service<br/>Redis-backed"] end subgraph INTEGRATION["Integration Layer"] LDAP_CLIENT["LDAP Client<br/>DirectoryService"] STORAGE_CLIENT["Storage Client<br/>Supabase SDK / S3 Flysystem"] EXPORT["Export Engine<br/>PDF · Excel · CSV"] QR_SVC["QR Code Service<br/>Document Verification"] end PHP_FPM --> KERNEL KERNEL --> ROUTES ROUTES --> MIDDLEWARE MIDDLEWARE --> APP_CORE APP_CORE --> WORKFLOW_NOTIF APP_CORE --> BACKGROUND APP_CORE --> INTEGRATION WORKFLOW_NOTIF --> BACKGROUND end %% ═══════════════════════════════════════════════════════════════ %% LAYER 4: DATA TIER %% ═══════════════════════════════════════════════════════════════ subgraph DATA["■■ DATA LAYER — Persistence Tier"] direction LR subgraph POSTGRES["PostgreSQL 17 — Primary Database"] PG_MAIN["simmak_ppkp<br/>Single Centralized Database"] PG_SCHEMA["Schemas<br/>public · audit_log"] PG_FEATURES["Features<br/>JSONB · UUID · Full-Text Search<br/>Row-Level Audit · Partitioning"] PG_BACKUP["pg_dump + WAL Archiving<br/>Daily Backup · PITR"] end subgraph REDIS_LAYER["Redis 7 — Cache &
```

```
Queue"] REDIS_CACHE["Cache Store<br/>Config · RBAC · Dashboard KPIs"] REDIS_SESSION["Session
Store<br/>Distributed Sessions"] REDIS_QUEUE["Queue Backend<br/>notifications · reports · ldap_sync"]
REDIS_LOCK["Distributed Locks<br/>Workflow Concurrency"] end end %%
========================================================================== %% LAYER 5: EXTERNAL SERVICES %%
========================================================================== subgraph EXTERNAL["■ EXTERNAL
SERVICES — Integration Tier"] direction TB subgraph LDAP_AD["LDAP / Active Directory"] AD_SERVER["AD
Pemprov DKI Jakarta<br/>ldap:// / ldaps://"] AD_AUTH["Authentication<br/>Bind + Search User"]
AD_SYNC["User Provisioning Sync<br/>Scheduled + On-Demand"] AD_GROUPS["Group → Role Mapping<br/>OU-based
Clinic Scope"] end subgraph SUPABASE["Supabase Storage — Object Storage"] SB_BUCKET["Buckets<br/>documents
· evidence · reports · attachments"] SB_RLS["Row-Level Security Policies"] SB_CDN["CDN Edge
Delivery<br/>Signed URLs · TTL"] SB_BACKUP["Cross-Region Replication<br/>(DR Strategy)"] end subgraph
OPTIONAL["Optional Integrations"] SMTP["SMTP / Mail Gateway<br/>Notification Delivery"]
MONITOR["Monitoring<br/>Prometheus · Grafana · Sentry"] end end %%
========================================================================== %% LAYER 6: INFRASTRUCTURE —
DOCKER COMPOSE %% ========================================================================== subgraph
INFRA["■ INFRASTRUCTURE — Docker Compose on Ubuntu Server"] direction LR DOCKER_NGINX["Container: nginx"]
DOCKER_APP["Container: app<br/>(PHP-FPM + Laravel)"] DOCKER_WORKER["Container: worker<br/>(Queue +
Scheduler)"] DOCKER_PG["Container: postgres:17"] DOCKER_REDIS["Container: redis:7-alpine"]
DOCKER_VOL["Volumes<br/>pg_data · redis_data · app_storage"] end %%
========================================================================== %% PRIMARY DATA FLOWS %%
========================================================================== CLIENT -->|"HTTPS :443"|
NGINX_LAYER NGINX_LAYER -->|"FastCGI / Proxy Pass"| ENTRY REPOS -->|"SQL Queries<br/>Read/Write"| POSTGRES
CACHE_SVC -->|"GET/SET/PUBLISH"| REDIS_LAYER QUEUE_WORKER -->|"BRPOP/LPUSH"| REDIS_LAYER BACKGROUND
-->|"Persist Job Results"| POSTGRES LDAP_CLIENT -->|"LDAP Bind / Search<br/>Port 389/636"| LDAP_AD
STORAGE_CLIENT -->|"REST API / S3 Protocol<br/>Upload · Download · Delete"| SUPABASE NOTIF_SVC -->|"SMTP
TLS"| SMTP DOCKER_NGINX --- NGINX DOCKER_APP --- PHP_FPM DOCKER_WORKER --- QUEUE_WORKER DOCKER_PG ---
PG_MAIN DOCKER_REDIS --- REDIS_CACHE %%
========================================================================== %% SECURITY BOUNDARIES %%
========================================================================== subgraph SECURITY["■ SECURITY
ARCHITECTURE"] direction LR SEC_WAF["WAF Rules<br/>(Nginx ModSecurity optional)"] SEC_JWT["API Token
Auth<br/>Sanctum Personal Access Tokens"] SEC_ENCRYPT["Encryption at Rest<br/>PG TDE · Supabase AES-256"]
SEC_TRANSIT["Encryption in Transit<br/>TLS 1.3 Everywhere"] SEC_AUDIT["Immutable Audit
Trail<br/>audit_trails · activity_logs"] end NGINX --> SEC_WAF MW_AUTH --> SEC_JWT POSTGRES -->
SEC_ENCRYPT SSL --> SEC_TRANSIT MW_AUDIT --> SEC_AUDIT %%
========================================================================== %% DOMAIN MODULE → DATA STORE
MAPPING %% ========================================================================== subgraph DATA_FLOW["■
Module Data Flow Summary"] direction TB DF1["DMS → PostgreSQL (metadata) + Supabase (files) + Workflow
Engine"] DF2["Audit/CAPA/Risk → PostgreSQL + Workflow + Notification"] DF3["Accreditation → PostgreSQL
(dynamic standards) + Supabase (evidence)"] DF4["Quality → PostgreSQL + Redis Cache (KPI dashboards)"]
DF5["Auth → PostgreSQL (users/RBAC) + LDAP/AD + Redis (sessions)"] end CLIENT_MODULES -.->|"Business
Operations"| DATA_FLOW DATA_FLOW -.-> DATA %%
========================================================================== %% STYLING — Layer Colors %%
========================================================================== classDef clientStyle
fill:#EBF8FF,stroke:#3182CE,stroke-width:2px,color:#1A365D classDef nginxStyle
fill:#FFF5EB,stroke:#DD6B20,stroke-width:2px,color:#7B341E classDef laravelStyle
fill:#F0FFF4,stroke:#38A169,stroke-width:2px,color:#22543D classDef dataStyle
fill:#FAF5FF,stroke:#805AD5,stroke-width:2px,color:#44337A classDef externalStyle
fill:#FFF5F5,stroke:#E53E3E,stroke-width:2px,color:#742A2A classDef infraStyle
fill:#EDF2F7,stroke:#4A5568,stroke-width:2px,color:#2D3748 classDef securityStyle
fill:#FFFFF0,stroke:#D69E2E,stroke-width:2px,color:#744210 classDef flowStyle
fill:#E6FFFA,stroke:#319795,stroke-width:1px,color:#234E52 class
BROWSER,SPA,PWA,EXEC_DASH,CLINIC_DASH,ACCRED_DASH,MOD_DMS,MOD_AUDIT,MOD_CAPA,MOD_RISK,MOD_ACCRED,MOD_QUALITY,MOD_ADMIN
clientStyle class NGINX,SSL,GZIP,RATE,STATIC,UPSTREAM nginxStyle class
PHP_FPM,KERNEL,ROUTES,MW_AUTH,MW_RBAC,MW_SCOPE,MW_AUDIT,MW_THROTTLE,CONTROLLERS,SERVICES,REPOS,EVENTS,POLICIES,WF_ENGINE
laravelStyle class
PG_MAIN,PG_SCHEMA,PG_FEATURES,PG_BACKUP,REDIS_CACHE,REDIS_SESSION,REDIS_QUEUE,REDIS_LOCK dataStyle class
AD_SERVER,AD_AUTH,AD_SYNC,AD_GROUPS,SB_BUCKET,SB_RLS,SB_CDN,SB_BACKUP,SMTP,MONITOR externalStyle class
DOCKER_NGINX,DOCKER_APP,DOCKER_WORKER,DOCKER_PG,DOCKER_REDIS,DOCKER_VOL infraStyle class
SEC_WAF,SEC_JWT,SEC_ENCRYPT,SEC_TRANSIT,SEC_AUDIT securityStyle class DF1,DF2,DF3,DF4,DF5 flowStyle %%
Subgraph styling style CLIENT fill:#EBF8FF,stroke:#3182CE,stroke-width:3px style NGINX_LAYER
fill:#FFF5EB,stroke:#DD6B20,stroke-width:3px style LARAVEL fill:#F0FFF4,stroke:#38A169,stroke-width:3px
style DATA fill:#FAF5FF,stroke:#805AD5,stroke-width:3px style POSTGRES
fill:#EDE9FE,stroke:#7C3AED,stroke-width:2px style REDIS_LAYER
fill:#FEE2E2,stroke:#DC2626,stroke-width:2px style EXTERNAL fill:#FFF5F5,stroke:#E53E3E,stroke-width:3px
style LDAP_AD fill:#FEF3C7,stroke:#D97706,stroke-width:2px style SUPABASE
```

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style SECURITY fill:#FFFFF0,stroke:#D69E2E,stroke-width:3px style DATA_FLOW
fill:#E6FFFA,stroke:#319795,stroke-width:2px
```

BPMN

[Diagram Mermaid — lihat file diagrams/*.mmd untuk versi visual]

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%% ============================================================================= %% SIMMAK PPKP —
BPMN-Style Process Flow Diagrams %% Sistem Informasi Manajemen Mutu, Audit Internal dan Akreditasi %% PPKP
Provinsi DKI Jakarta %% Syntax: flowchart with swimlanes (subgraphs) | BPMN-inspired notation %%
============================================================================= %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 1 — DOCUMENT WORKFLOW (Draft → Review → Approve → Publish) %% Configurable via Workflow Engine |
Default: Draft → Reviewer → PJ Mutu → Kepala Klinik → Published %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_DOC["■ DOCUMENT WORKFLOW — DMS Module"] direction TB START_DOC(("■ Start"))
subgraph LANE_AUTHOR["Swimlane: Document Author"] direction TB A1["Create Document<br/>(Kebijakan / SOP /
Pedoman /<br/>Formulir / Instruksi Kerja)"] A2["Edit Draft Content<br/>& Attachments"]
A3{"Draft<br/>Complete?"} A4["Revise per<br/>Reviewer Comments"] A5["Revise per<br/>Approver Rejection"]
end subgraph LANE_REVIEWER["Swimlane: Reviewer"] direction TB R1["Receive Review<br/>Task Notification"]
R2{"Review<br/>Decision"} end subgraph LANE_PJ["Swimlane: PJ Mutu"] direction TB P1["Receive
Approval<br/>Task"] P2{"Quality<br/>Approval?"} end subgraph LANE_KK["Swimlane: Kepala Klinik"] direction
TB K1["Final Document<br/>Approval"] K2{"Approve for<br/>Publication?"} end subgraph
LANE_SYSTEM["Swimlane: System / Workflow Engine"] direction TB S1["Assign Workflow Steps<br/>(Dynamic
Configuration)"] S2["Increment Version Number"] S3["Generate QR Verification Code"] S4["Set Expiry Date
&<br/>Schedule Monitoring"] S5["Publish to Document Repository"] S6["Send Distribution<br/>Notification"]
S7["Log Audit Trail"] end END_PUB(("■ Published")) END_REJ(("■ Rejected /<br/>Archived")) START_DOC --> A1
A1 --> A2 --> A3 A3 -->|"No"| A2 A3 -->|"Yes"| S1 S1 --> R1 --> R2 R2 -->|"Request Revision"| A4 --> A2 R2
-->|"Approved"| P1 R2 -->|"Rejected"| END_REJ P1 --> P2 P2 -->|"Revision Required"| A5 --> A2 P2
-->|"Approved"| K1 K1 --> K2 K2 -->|"Rejected"| A5 K2 -->|"Approved"| S2 S2 --> S3 --> S4 --> S5 --> S6
--> S7 --> END_PUB end classDef startend fill:#4CAF50,stroke:#2E7D32,stroke-width:3px,color:#fff classDef
task fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef gateway
fill:#FFE082,stroke:#F57F17,stroke-width:2px classDef system fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
class START_DOC,END_PUB startend class END_REJ fill:#EF5350,stroke:#C62828,stroke-width:3px,color:#fff
class A1,A2,A4,A5,R1,P1,K1 task class A3,R2,P2,K2 gateway class S1,S2,S3,S4,S5,S6,S7 system --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 2 — INTERNAL AUDIT LIFECYCLE %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_AUD["■ INTERNAL AUDIT LIFECYCLE"] direction TB START_AUD(("■ Start")) subgraph
LANE_MGT["Swimlane: Manajemen Pusat / Audit Manager"] direction TB M1["Define Annual<br/>Audit Program"]
M2["Approve Audit<br/>Schedule"] M3["Review Draft<br/>Audit Report"] M4{"Report<br/>Approved?"}
M5["Approve Audit<br/>Closure"] end subgraph LANE_LEAD["Swimlane: Lead Auditor"] direction TB L1["Create
Audit Schedule<br/>(Clinic / Unit / Scope)"] L2["Assign Audit Team<br/>& Lead Auditor"] L3["Prepare Audit
Checklist<br/>(from Template)"] L4["Conduct Opening<br/>Meeting"] L6["Compile Audit<br/>Findings"]
L7["Draft Audit Report"] L8["Conduct Closing<br/>Meeting"] end subgraph LANE_TEAM["Swimlane: Auditor
Team"] direction TB T1["Execute Fieldwork<br/>& Checklist"] T2["Collect Evidence<br/>(Photos, Documents)"]
T3["Document Observations<br/>& Non-Conformities"] end subgraph LANE_AUDITEE["Swimlane: Auditee / Klinik"]
direction TB D1["Receive Audit<br/>Notification"] D2["Provide Documents<br/>& Access"] D3["Receive
Findings<br/>& Report"] D4["Acknowledge<br/>Audit Results"] end subgraph LANE_SYS_AUD["Swimlane: System"]
direction TB SA1["Register Findings<br/>in Finding Register"] SA2["Auto-Create CAPA<br/>for Each Major
NC"] SA3["Update Audit Status<br/>& Dashboard Metrics"] end END_AUD_OK(("■ Audit Closed")) END_AUD_REV(("■
Report Revision")) START_AUD --> M1 --> L1 --> M2 M2 --> L2 --> L3 --> D1 D1 --> L4 --> D2 --> T1 T1 -->
T2 --> T3 --> L6 L6 --> SA1 --> L7 --> M3 --> M4 M4 -->|"Revision"| END_AUD_REV --> L7 M4 -->|"Approved"|
L8 --> D3 --> D4 D4 --> M5 --> SA2 --> SA3 --> END_AUD_OK end classDef startend
fill:#4CAF50,stroke:#2E7D32,stroke-width:3px,color:#fff classDef mgmt
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef lead fill:#F3E5F5,stroke:#6A1B9A,stroke-width:2px
classDef team fill:#EDE7F6,stroke:#4527A0,stroke-width:2px classDef auditee
fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef sys fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
classDef gw fill:#FFE082,stroke:#F57F17,stroke-width:2px class START_AUD,END_AUD_OK startend class
M1,M2,M3,M5 mgmt class L1,L2,L3,L4,L6,L7,L8 lead class T1,T2,T3 team class D1,D2,D3,D4 auditee class
SA1,SA2,SA3 sys class M4 gw --- %%
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■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 3 — CAPA LIFECYCLE %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_CAPA["■■ CAPA LIFECYCLE — Corrective & Preventive Action"] direction TB
START_CAPA(("■ Start")) subgraph LANE_SRC["Swimlane: Finding Source"] direction TB FS1["Audit Finding
/<br/>Risk Escalation /<br/>Quality Issue"] end subgraph LANE_COORD["Swimlane: PJ Mutu / CAPA
Coordinator"] direction TB C1["Register CAPA Record"] C2["Assign CAPA Owner<br/>& Set Due Date"]
C3["Monitor CAPA<br/>Progress"] C4{"Verification<br/>Passed?"} C5{"Effectiveness<br/>Review OK?"}
C6["Approve CAPA<br/>Closure"] end subgraph LANE_OWNER["Swimlane: CAPA Owner / PIC"] direction TB
O1["Perform Root Cause Analysis"] O2{"RCA Method?"} O3["5 Why Analysis"] O4["Fishbone Diagram"] O5["Define
Corrective<br/>Action Plan"] O6["Define Preventive<br/>Action Plan"] O7["Implement Actions<br/>& Upload
Evidence"] end subgraph LANE_VER["Swimlane: Verifier / QA"] direction TB V1["Verify Implementation<br/>of
Actions"] V2["Conduct Effectiveness<br/>Review (30/60/90 days)"] end subgraph LANE_KK_CAPA["Swimlane:
Kepala Klinik"] direction TB KK1["Approve CAPA<br/>Assignment"] KK2["Sign-off CAPA<br/>Closure"] end
subgraph LANE_SYS_CAPA["Swimlane: System"] direction TB SC1["Link to Source Finding"] SC2["Send Deadline
Reminders"] SC3["Update CAPA Dashboard<br/>& Open Findings Count"] end END_CAPA_OK(("■ CAPA Closed"))
END_CAPA_REOPEN(("■ CAPA Reopened")) START_CAPA --> FS1 --> C1 --> SC1 --> KK1 KK1 --> C2 --> O1 --> O2 O2
-->|"5 Why"| O3 --> O5 O2 -->|"Fishbone"| O4 --> O5 O5 --> O6 --> O7 O7 --> SC2 --> V1 --> C4 C4
-->|"Failed"| END_CAPA_REOPEN --> O1 C4 -->|"Passed"| V2 --> C5 C5 -->|"Not Effective"| END_CAPA_REOPEN C5
-->|"Effective"| C6 --> KK2 --> SC3 --> END_CAPA_OK end classDef startend
fill:#4CAF50,stroke:#2E7D32,stroke-width:3px,color:#fff classDef coord
fill:#FCE4EC,stroke:#C2185B,stroke-width:2px classDef owner fill:#FFF3E0,stroke:#E65100,stroke-width:2px
classDef ver fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef kk
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef sys fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
classDef gw fill:#FFE082,stroke:#F57F17,stroke-width:2px class START_CAPA,END_CAPA_OK startend class FS1
coord class C1,C2,C3,C6 coord class O1,O3,O4,O5,O6,O7 owner class V1,V2 ver class KK1,KK2 kk class
SC1,SC2,SC3 sys class C4,C5,O2 gw class END_CAPA_REOPEN
fill:#EF5350,stroke:#C62828,stroke-width:3px,color:#fff --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 4 — RISK ASSESSMENT %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_RISK["■■ RISK ASSESSMENT PROCESS"] direction TB START_RISK(("■ Start")) subgraph
LANE_IDENT["Swimlane: Risk Identifier (Staff / Unit)"] direction TB I1["Identify Risk Event<br/>(Hazard /
Threat / Opportunity)"] I2["Describe Risk Context<br/>& Category"] end subgraph LANE_OWNER_R["Swimlane:
Risk Owner"] direction TB RO1["Register Risk<br/>in Risk Register"] RO2["Assess Likelihood<br/>(1–5
Scale)"] RO3["Assess Impact<br/>(1–5 Scale)"] RO4{"Inherent Risk<br/>Level?"} RO5["Define
Mitigation<br/>Actions & Timeline"] RO6["Implement Mitigation<br/>Controls"] RO7["Re-assess
Residual<br/>Risk Score"] end subgraph LANE_COMM["Swimlane: Risk Committee / Manajemen Pusat"] direction
TB RC1["Review High & Extreme<br/>Risks"] RC2{"Accept Residual<br/>Risk?"} RC3["Escalate to CAPA"]
RC4["Approve Risk<br/>Acceptance"] end subgraph LANE_PJ_R["Swimlane: PJ Mutu"] direction TB PJ1["Validate
Risk<br/>Assessment Method"] PJ2["Monitor Risk Register<br/>& Heatmap"] end subgraph LANE_SYS_R["Swimlane:
System"] direction TB SR1["Calculate Risk Score<br/>(L × I)"] SR2["Plot on Risk Matrix<br/>& Heatmap"]
SR3["Schedule Periodic<br/>Risk Review"] SR4["Update Executive<br/>Dashboard"] end END_RISK_MON(("■ Under
Monitoring")) END_RISK_CAPA(("■ Escalated to CAPA")) START_RISK --> I1 --> I2 --> RO1 RO1 --> PJ1 --> RO2
--> RO3 --> SR1 --> RO4 RO4 -->|"Low / Medium"| RO5 RO4 -->|"High / Extreme"| RC1 RC1 --> RO5 RO5 --> RO6
--> RO7 --> SR2 --> RC2 RC2 -->|"Not Acceptable"| RC3 --> END_RISK_CAPA RC2 -->|"Acceptable"| RC4 --> SR3
--> PJ2 --> SR4 --> END_RISK_MON end classDef startend
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fill:#FFEBEE,stroke:#B71C1C,stroke-width:2px classDef owner fill:#FFF3E0,stroke:#E65100,stroke-width:2px
classDef comm fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef pj
fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef sys fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
classDef gw fill:#FFE082,stroke:#F57F17,stroke-width:2px class START_RISK,END_RISK_MON startend class
I1,I2 ident class RO1,RO2,RO3,RO5,RO6,RO7 owner class RC1,RC3,RC4 comm class PJ1,PJ2 pj class
SR1,SR2,SR3,SR4 sys class RO4,RC2 gw class END_RISK_CAPA
fill:#EF5350,stroke:#C62828,stroke-width:3px,color:#fff --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 5 — ACCREDITATION SELF ASSESSMENT %% Supports FKTP & FKTL | Dynamic Standard Hierarchy from UI %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_SA["■ ACCREDITATION SELF ASSESSMENT"] direction TB START_SA(("■ Start")) subgraph
LANE_ADMIN["Swimlane: Accreditation Admin"] direction TB AD1["Select Accreditation Type<br/>(FKTP /
FKTL)"] AD2["Configure Assessment Period<br/>& Target Clinic"] AD3["Activate Self
Assessment<br/>Campaign"] end subgraph LANE_STD["Swimlane: System — Standard Engine"] direction TB
ST1["Load Standard Hierarchy<br/>Bab → Standar → Kriteria<br/>→ Elemen Penilaian"] ST2["Generate
Assessment<br/>Checklist per Clinic"] end subgraph LANE_ASSESSOR["Swimlane: Clinic Assessor / PJ Mutu
Klinik"] direction TB AS1["Receive Self Assessment<br/>Assignment"] AS2["Evaluate Each<br/>Elemen
```

```
Penilaian"] AS3{"Compliance<br/>Status?"} AS4["Mark: Ya (Comply)"] AS5["Mark: Tidak (Non-Comply)"]
AS6["Mark: N/A"] AS7["Attach Evidence<br/>per Element"] AS8{"All Elements<br/>Completed?"} AS9["Submit
Self Assessment<br/>for Review"] end subgraph LANE_STAFF["Swimlane: Unit Staff"] direction TB US1["Provide
Supporting<br/>Documents & Data"] US2["Upload Evidence<br/>to Repository"] end subgraph
LANE_KK_SA["Swimlane: Kepala Klinik"] direction TB KKSA1["Review Self Assessment<br/>Results"]
KKSA2{"Approve<br/>Submission?"} KKSA3["Sign-off Self<br/>Assessment Report"] end subgraph
LANE_SYS_SA["Swimlane: System"] direction TB SSA1["Calculate Compliance %<br/>per Kriteria / Standar /
Bab"] SSA2["Identify Non-Compliant<br/>Elements for Gap Analysis"] SSA3["Update Readiness<br/>Dashboard"]
end END_SA(("■ Self Assessment Complete")) END_SA_REV(("■ Returned for Revision")) START_SA --> AD1 -->
AD2 --> AD3 --> ST1 --> ST2 --> AS1 AS1 --> AS2 --> AS3 AS3 -->|"Comply"| AS4 --> AS7 AS3
-->|"Non-Comply"| AS5 --> AS7 AS3 -->|"Not Applicable"| AS6 --> AS8 AS7 --> US1 --> US2 --> AS8 AS8
-->|"No"| AS2 AS8 -->|"Yes"| AS9 --> KKSA1 --> KKSA2 KKSA2 -->|"Revision"| END_SA_REV --> AS2 KKSA2
-->|"Approved"| KKSA3 --> SSA1 --> SSA2 --> SSA3 --> END_SA end classDef startend
fill:#4CAF50,stroke:#2E7D32,stroke-width:3px,color:#fff classDef admin
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef std fill:#EDE7F6,stroke:#4527A0,stroke-width:2px
classDef assess fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef staff
fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef kk fill:#F1F8E9,stroke:#558B2F,stroke-width:2px
classDef sys fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px classDef gw
fill:#FFE082,stroke:#F57F17,stroke-width:2px class START_SA,END_SA startend class AD1,AD2,AD3 admin class
ST1,ST2 std class AS1,AS2,AS4,AS5,AS6,AS7,AS9 assess class US1,US2 staff class KKSA1,KKSA3 kk class
SSA1,SSA2,SSA3 sys class AS3,AS8,KKSA2 gw class END_SA_REV
fill:#FF9800,stroke:#E65100,stroke-width:3px,color:#fff --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS 6 — GAP ANALYSIS %% Triggered after Self Assessment | Links to CAPA & Evidence Repository %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph SW_GAP["■ GAP ANALYSIS PROCESS"] direction TB START_GAP(("■ Start")) subgraph
LANE_TRIGGER["Swimlane: System Trigger"] direction TB GT1["Receive Non-Compliant<br/>Elements
from<br/>Self Assessment"] GT2["Auto-Generate<br/>Gap Records"] end subgraph LANE_PJ_GAP["Swimlane: PJ
Mutu / Clinic Quality Officer"] direction TB G1["Review Identified Gaps<br/>per Standar & Kriteria"]
G2["Categorize Gap Type<br/>(Documentation / Process /<br/>Infrastructure / Competency)"] G3["Determine
Root Cause<br/>of Each Gap"] G4["Assign Priority<br/>(Critical / Major / Minor)"] G5["Define
Remediation<br/>Action Plan"] G6["Set Target Date<br/>& Responsible PIC"] G7{"Gap
Remediation<br/>Completed?"} G8["Verify Closure<br/>Evidence"] end subgraph LANE_PIC["Swimlane: PIC / Unit
Owner"] direction TB P1["Execute Remediation<br/>Actions"] P2["Upload Closure<br/>Evidence"] P3["Link DMS
Documents<br/>to Gap Record"] end subgraph LANE_MGT_GAP["Swimlane: Manajemen Pusat"] direction TB
MG1["Review Critical & Major<br/>Gaps Across 17 Clinics"] MG2["Approve Remediation<br/>Plan (if
Critical)"] MG3["Monitor Clinic<br/>Gap Closure Rate"] end subgraph LANE_SYS_GAP["Swimlane: System"]
direction TB SG1["Calculate Gap Score<br/>& Readiness Impact"] SG2{"Critical Gap<br/>Unresolved?"}
SG3["Auto-Create CAPA<br/>from Critical Gap"] SG4["Update Gap Dashboard<br/>& Clinic Ranking"]
SG5["Generate Gap Analysis<br/>Report (PDF)"] end END_GAP_OK(("■ Gap Closed")) END_GAP_CAPA(("■ CAPA
Created")) START_GAP --> GT1 --> GT2 --> G1 --> G2 --> G3 --> G4 --> G5 --> G6 G6 --> MG1 --> MG2 --> P1
P1 --> P2 --> P3 --> G7 G7 -->|"No"| P1 G7 -->|"Yes"| G8 --> SG1 --> SG2 SG2 -->|"Yes, Overdue"| SG3 -->
END_GAP_CAPA SG2 -->|"No / Resolved"| SG4 --> MG3 --> SG5 --> END_GAP_OK end classDef startend
fill:#4CAF50,stroke:#2E7D32,stroke-width:3px,color:#fff classDef trigger
fill:#EDE7F6,stroke:#4527A0,stroke-width:2px classDef pj fill:#E0F7FA,stroke:#00695C,stroke-width:2px
classDef pic fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef mgmt
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef sys fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
classDef gw fill:#FFE082,stroke:#F57F17,stroke-width:2px class START_GAP,END_GAP_OK startend class GT1,GT2
trigger class G1,G2,G3,G4,G5,G6,G8 pj class P1,P2,P3 pic class MG1,MG2,MG3 mgmt class SG1,SG2,SG3,SG4,SG5
sys class G7,SG2 gw class END_GAP_CAPA fill:#EF5350,stroke:#C62828,stroke-width:3px,color:#fff --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %%
PROCESS OVERVIEW — CROSS-MODULE INTEGRATION MAP %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart LR subgraph OVERVIEW["SIMMAK PPKP — Process Integration Overview"] direction LR DOC_WF["■
Document Workflow<br/>Draft → Review → Approve → Publish"] AUD_LC["■ Internal Audit<br/>Lifecycle"]
CAPA_LC["■■ CAPA<br/>Lifecycle"] RISK_P["■■ Risk Assessment"] SA_P["■ Self Assessment"] GAP_P["■ Gap
Analysis"] DOC_WF -->|"Controlled Documents<br/>as Audit Evidence"| AUD_LC AUD_LC -->|"Findings"| CAPA_LC
RISK_P -->|"High/Extreme Risk"| CAPA_LC SA_P -->|"Non-Compliant Elements"| GAP_P GAP_P -->|"Critical
Gaps"| CAPA_LC GAP_P -->|"Evidence Links"| DOC_WF CAPA_LC -->|"Effectiveness Data"| RISK_P CAPA_LC
-->|"Closure Updates"| SA_P end classDef p1 fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef p2
fill:#F3E5F5,stroke:#6A1B9A,stroke-width:2px classDef p3 fill:#FCE4EC,stroke:#C2185B,stroke-width:2px
classDef p4 fill:#FFEBEE,stroke:#B71C1C,stroke-width:2px classDef p5
fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef p6 fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px
class DOC_WF p1 class AUD_LC p2 class CAPA_LC p3 class RISK_P p4 class SA_P p5 class GAP_P p6
```

DFD

[Diagram Mermaid — lihat file diagrams/*.mmd untuk versi visual]

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%% ============================================================================== %% SIMMAK PPKP — Data
Flow Diagrams (DFD) %% Sistem Informasi Manajemen Mutu, Audit Internal dan Akreditasi %% PPKP Provinsi DKI
Jakarta %% Syntax: flowchart with subgraphs | Notation: Yourdon-DeMarco style %%
============================================================================== %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %% DFD
LEVEL 0 — CONTEXT DIAGRAM %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph EXT["External Entities"] direction TB E1["■ Manajemen Pusat<br/>(Executive /
Direksi)"] E2["■ Kepala Klinik<br/>(17 Entitas Klinik)"] E3["■ PJ Mutu / Quality Officer"] E4["■ Auditor
Internal"] E5["■ Staff Klinik / Unit Kerja"] E6["■ Active Directory<br/>Pemprov DKI Jakarta"] E7["■
Supabase / S3<br/>Object Storage"] end subgraph SYS["Process 0 — SIMMAK PPKP<br/>Sistem Informasi
Manajemen Mutu,<br/>Audit Internal dan Akreditasi"] direction TB P0["0<br/>SIMMAK PPKP<br/>Integrated
Quality Platform"] end E1 -->|"KPI, Readiness Report,<br/>Executive Dashboard"| P0 P0 -->|"Accreditation
Readiness,<br/>Risk Heatmap, Clinic Ranking"| E1 E2 -->|"Approval Dokumen,<br/>Audit Schedule, CAPA
Assignment"| P0 P0 -->|"Clinic Dashboard,<br/>Open Findings, Expired Docs"| E2 E3 -->|"Document
Draft/Review,<br/>Quality Program, Indicators"| P0 P0 -->|"Workflow Tasks, CAPA Status,<br/>Quality
Monitoring Alerts"| E3 E4 -->|"Audit Program, Checklist,<br/>Findings, Audit Report"| P0 P0 -->|"Audit
Schedule, Finding Register,<br/>CAPA Trigger"| E4 E5 -->|"Self Assessment Input,<br/>Evidence Upload, Risk
Input"| P0 P0 -->|"Notifications, Document Access,<br/>Task Assignments"| E5 E6 -->|"LDAP
Authentication<br/>User Credentials"| P0 P0 -->|"User Sync / SSO Token<br/>Validation Request"| E6 P0
<-->|"Document Files,<br/>Evidence Attachments"| E7 classDef external
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px,color:#0D47A1 classDef system
fill:#E8F5E9,stroke:#2E7D32,stroke-width:3px,color:#1B5E20 class E1,E2,E3,E4,E5,E6,E7 external class P0
system --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %% DFD
LEVEL 1 — SYSTEM DECOMPOSITION (ALL MODULES) %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph ENT["External Entities"] direction LR U1["Users / Staff"] U2["LDAP / AD"] U3["Object
Storage"] end subgraph CORE["1.0 Core Platform"] direction TB P1["1.1 Authentication<br/>& Session Mgmt"]
P2["1.2 RBAC &<br/>Permission Engine"] P3["1.3 Workflow<br/>Engine"] P4["1.4 Notification<br/>Service"]
P5["1.5 Audit Trail &<br/>Activity Log"] D1[("D1: users<br/>roles<br/>permissions")] D2[("D2:
workflow_defs<br/>workflow_instances")] D3[("D3: audit_logs<br/>activity_logs")] end subgraph DMS["2.0
Document Management System"] direction TB P6["2.1 Document<br/>Lifecycle Mgmt"] P7["2.2 Version
Control<br/>& QR Verify"] P8["2.3 Expiry<br/>Monitoring"] D4[("D4:
documents<br/>document_versions<br/>document_types")] end subgraph AUD["3.0 Internal Audit"] direction TB
P9["3.1 Audit Program<br/>& Schedule"] P10["3.2 Checklist<br/>Execution"] P11["3.3
Finding<br/>Registration"] D5[("D5: audit_programs<br/>audit_schedules<br/>audit_findings")] end subgraph
CAPA["4.0 CAPA Management"] direction TB P12["4.1 Root Cause<br/>Analysis"] P13["4.2 Corrective
&<br/>Preventive Action"] P14["4.3 Verification &<br/>Effectiveness Review"] D6[("D6:
capa_records<br/>rca_analyses<br/>capa_actions")] end subgraph RISK["5.0 Risk Management"] direction TB
P15["5.1 Risk Register<br/>& Assessment"] P16["5.2 Risk Matrix<br/>& Heatmap"] P17["5.3 Mitigation
Plan<br/>& Monitoring"] D7[("D7: risks<br/>risk_assessments<br/>mitigation_plans")] end subgraph ACC["6.0
Accreditation Management"] direction TB P18["6.1 Standard<br/>Configuration"] P19["6.2 Self
Assessment<br/>& Mock Survey"] P20["6.3 Gap Analysis<br/>& Evidence Repo"] D8[("D8:
standards<br/>criteria<br/>assessments<br/>gaps")] end subgraph QMS["7.0 Quality Management"] direction TB
P21["7.1 Quality Program<br/>& Indicators"] P22["7.2 Quality Monitoring<br/>& Evaluation"] D9[("D9:
quality_programs<br/>quality_indicators<br/>evaluations")] end subgraph DASH["8.0 Dashboard & Reporting"]
direction TB P23["8.1 Executive<br/>Dashboard"] P24["8.2 Clinic<br/>Dashboard"] P25["8.3
Report<br/>Generator"] D10[("D10: dashboard_cache<br/>report_snapshots")] end subgraph ORG["9.0
Organization Scope"] D11[("D11: organizations<br/>clinics<br/>units")] end U1 --> P1 & P6 & P9 & P12 & P15
& P18 & P21 & P23 U2 <-->|"LDAP Auth"| P1 P1 --> D1 P1 --> P2 P2 --> D1 P2 --> P3 P3 --> D2 P3 --> P4 P4
--> U1 P5 --> D3 P6 --> D4 P6 --> P3 P7 --> D4 P7 --> U3 P8 --> D4 P8 --> P4 P9 --> D5 P10 --> D5 P11 -->
D5 P11 -->|"Finding Trigger"| P12 P12 --> D6 P13 --> D6 P13 --> P3 P14 --> D6 P14 --> P4 P15 --> D7 P16
--> D7 P17 --> D7 P17 --> P4 P18 --> D8 P19 --> D8 P19 --> U3 P20 --> D8 P20 --> D4 P21 --> D9 P22 --> D9
P23 --> D10 P24 --> D10 P25 --> D10 D4 & D5 & D6 & D7 & D8 & D9 --> P23 & P24 P1 & P6 & P9 & P12 & P15 &
P18 & P21 --> P5 D11 -.->|"organization_id<br/>clinic_id<br/>unit_id"| D1 & D4 & D5 & D6 & D7 & D8 & D9
classDef ext fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef core
fill:#E8F5E9,stroke:#2E7D32,stroke-width:2px classDef dms fill:#FFF3E0,stroke:#E65100,stroke-width:2px
classDef audit fill:#F3E5F5,stroke:#6A1B9A,stroke-width:2px classDef capa
```

```
fill:#FCE4EC,stroke:#C2185B,stroke-width:2px classDef risk fill:#FFEBEE,stroke:#B71C1C,stroke-width:2px
classDef acc fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef qms
fill:#F1F8E9,stroke:#558B2F,stroke-width:2px classDef dash fill:#EDE7F6,stroke:#4527A0,stroke-width:2px
classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px,color:#E65100 class U1,U2,U3 ext class
P1,P2,P3,P4,P5 core class P6,P7,P8 dms class P9,P10,P11 audit class P12,P13,P14 capa class P15,P16,P17
risk class P18,P19,P20 acc class P21,P22 qms class P23,P24,P25 dash class
D1,D2,D3,D4,D5,D6,D7,D8,D9,D10,D11 ds --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %% DFD
LEVEL 2 — DOCUMENT MANAGEMENT SYSTEM (DMS) %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph EXT_DMS["External Entities"] ED1["Document Author"] ED2["Reviewer"]
ED3["Approver<br/>(PJ Mutu / Kepala Klinik)"] ED4["End User / Staff"] ED5["Object Storage"] end subgraph
P_DMS["2.0 Document Management System"] direction TB P21["2.1.1 Create &<br/>Edit Document"] P22["2.1.2
Submit for<br/>Review Workflow"] P23["2.1.3 Review &<br/>Comment"] P24["2.1.4 Approve /<br/>Reject"]
P25["2.1.5 Publish &<br/>Distribute"] P26["2.2.1 Version<br/>Increment"] P27["2.2.2 Generate
QR<br/>Verification Code"] P28["2.3.1 Expiry Date<br/>Calculation"] P29["2.3.2 Expiry
Alert<br/>Notification"] end subgraph DS_DMS["Data Stores"] DD1[("documents")] DD2[("document_versions")]
DD3[("document_workflows")] DD4[("document_types<br/>Kebijakan/SOP/Pedoman<br/>Formulir/IK")]
DD5[("document_access_logs")] DD6[("expiry_alerts")] end ED1 -->|"Draft Content,<br/>Metadata,
Attachments"| P21 P21 --> DD1 & DD2 & DD4 P21 -->|"Upload File"| ED5 ED5 -->|"Stored File URL"| P21 P21
-->|"Submit Request"| P22 P22 --> DD3 P22 -->|"Workflow Task"| ED2 ED2 -->|"Review
Comments,<br/>Approve/Reject"| P23 P23 --> DD3 & DD2 P23 --> P24 ED3 -->|"Final Approval"| P24 P24 --> DD3
P24 -->|"Approved"| P25 P25 --> DD1 P25 -->|"Published Doc<br/>Notification"| ED4 P25 --> P26 P26 --> DD2
P26 --> P27 P27 --> DD1 P27 -->|"QR Code Label"| ED4 P25 --> P28 P28 --> DD1 & DD6 P28 --> P29 P29 --> DD6
P29 -->|"Expiry Warning<br/>30/60/90 days"| ED1 & ED3 ED4 -->|"QR Scan / Access Request"| P27 P27 --> DD5
classDef ext fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef proc
fill:#FFF3E0,stroke:#E65100,stroke-width:2px classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px
class ED1,ED2,ED3,ED4,ED5 ext class P21,P22,P23,P24,P25,P26,P27,P28,P29 proc class DD1,DD2,DD3,DD4,DD5,DD6
ds --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %% DFD
LEVEL 2 — INTERNAL AUDIT %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph EXT_AUD["External Entities"] EA1["Audit Manager /<br/>Manajemen Pusat"] EA2["Lead
Auditor"] EA3["Auditor Team"] EA4["Auditee /<br/>Unit Klinik"] EA5["Kepala Klinik"] end subgraph
P_AUD["3.0 Internal Audit Module"] direction TB P31["3.2.1 Define Audit<br/>Program (Annual)"] P32["3.2.2
Create Audit<br/>Schedule"] P33["3.2.3 Assign Audit<br/>Team & Scope"] P34["3.2.4 Prepare<br/>Checklist"]
P35["3.2.5 Conduct<br/>Audit (Fieldwork)"] P36["3.2.6 Record<br/>Findings"] P37["3.2.7 Draft
Audit<br/>Report"] P38["3.2.8 Review &<br/>Approve Report"] P39["3.2.9 Close Audit &<br/>Trigger CAPA"]
end subgraph DS_AUD["Data Stores"] DA1[("audit_programs")] DA2[("audit_schedules")] DA3[("audit_teams")]
DA4[("audit_checklists<br/>audit_checklist_items")]
DA5[("audit_findings<br/>observations<br/>non_conformities")] DA6[("audit_reports")]
DA7[("audit_evidence")] end EA1 -->|"Annual Audit Plan,<br/>Scope Definition"| P31 P31 --> DA1 P31 --> P32
EA1 -->|"Schedule Approval"| P32 P32 --> DA2 P32 -->|"Schedule Notification"| EA4 & EA5 P32 --> P33 EA2
-->|"Team Assignment"| P33 P33 --> DA3 P33 --> P34 P34 --> DA4 P34 -->|"Checklist Template"| EA3 EA3
-->|"Checklist Responses,<br/>Evidence Photos"| P35 P35 --> DA4 & DA7 P35 --> P36 EA3 -->|"Finding
Details,<br/>Severity, Clause Ref"| P36 P36 --> DA5 P36 --> P37 P37 --> DA6 P37 -->|"Draft Report"| EA2
P37 --> P38 EA1 & EA5 -->|"Report Approval"| P38 P38 --> DA6 P38 --> P39 P39 --> DA2 P39 -->|"CAPA
Request<br/>(per Finding)"| CAPA_PROC["4.0 CAPA Module"] P39 -->|"Audit Completion<br/>Notification"| EA4
classDef ext fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef proc
fill:#F3E5F5,stroke:#6A1B9A,stroke-width:2px classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px
classDef cross fill:#FCE4EC,stroke:#C2185B,stroke-width:2px class EA1,EA2,EA3,EA4,EA5 ext class
P31,P32,P33,P34,P35,P36,P37,P38,P39 proc class DA1,DA2,DA3,DA4,DA5,DA6,DA7 ds class CAPA_PROC cross --- %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ %% DFD
LEVEL 2 — CAPA MANAGEMENT %%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
flowchart TB subgraph EXT_CAPA["External Entities"] EC1["Finding Source<br/>(Audit / Risk / Quality)"]
EC2["CAPA Owner /<br/>PIC Unit"] EC3["Root Cause<br/>Analyst"] EC4["Verifier /<br/>PJ Mutu"] EC5["Kepala
Klinik"] end subgraph P_CAPA["4.0 CAPA Management"] direction TB P41["4.3.1 Register CAPA<br/>from
Finding"] P42["4.3.2 Assign CAPA<br/>Owner & Due Date"] P43["4.3.3 Root Cause<br/>Analysis"] P44["4.3.4
Define Corrective<br/>Action Plan"] P45["4.3.5 Define Preventive<br/>Action Plan"] P46["4.3.6
Implement<br/>Actions"] P47["4.3.7 Verification<br/>of Implementation"] P48["4.3.8
Effectiveness<br/>Review"] P49["4.3.9 Close / Reopen<br/>CAPA"] end subgraph DS_CAPA["Data Stores"]
DC1[("capa_records")] DC2[("rca_analyses<br/>5 Why / Fishbone")] DC3[("corrective_actions")]
DC4[("preventive_actions")] DC5[("capa_verifications")] DC6[("effectiveness_reviews")]
DC7[("capa_attachments")] end EC1 -->|"Finding ID, Severity,<br/>Source Reference"| P41 P41 --> DC1 P41
```

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--> P42 EC5 -->|"Owner Assignment<br/>Approval"| P42 P42 --> DC1 P42 -->|"CAPA
Assignment<br/>Notification"| EC2 EC2 --> P43 EC3 -->|"RCA Input<br/>(5 Why / Fishbone)"| P43 P43 --> DC2
P43 --> P44 & P45 EC2 -->|"Action Plan,<br/>Timeline, Resources"| P44 & P45 P44 --> DC3 P45 --> DC4 P44 &
P45 --> P46 EC2 -->|"Implementation Evidence,<br/>Progress Updates"| P46 P46 --> DC3 & DC4 & DC7 P46 -->
P47 EC4 -->|"Verification Result,<br/>Evidence Check"| P47 P47 --> DC5 P47 --> P48 EC4 -->|"Effectiveness
Assessment,<br/>Follow-up Period"| P48 P48 --> DC6 P48 --> P49 EC5 -->|"Close Approval"| P49 P49 --> DC1
P49 -->|"CAPA Closed<br/>Notification"| EC1 & EC2 P49 -->|"Reopen Request"| P43 classDef ext
fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef proc fill:#FCE4EC,stroke:#C2185B,stroke-width:2px
classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px class EC1,EC2,EC3,EC4,EC5 ext class
P41,P42,P43,P44,P45,P46,P47,P48,P49 proc class DC1,DC2,DC3,DC4,DC5,DC6,DC7 ds --- %%
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ %% DFD
LEVEL 2 — RISK MANAGEMENT %%
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
flowchart TB subgraph EXT_RISK["External Entities"] ER1["Risk Owner /<br/>Unit Head"] ER2["Risk Committee
/<br/>Manajemen Pusat"] ER3["PJ Mutu"] ER4["All Clinic Staff"] end subgraph P_RISK["5.0 Risk Management"]
direction TB P51["5.4.1 Identify &<br/>Register Risk"] P52["5.4.2 Risk Assessment<br/>(Likelihood ×
Impact)"] P53["5.4.3 Risk Matrix<br/>Classification"] P54["5.4.4 Define Mitigation<br/>Plan"] P55["5.4.5
Implement<br/>Mitigation"] P56["5.4.6 Risk Monitoring<br/>& Review"] P57["5.4.7 Generate
Risk<br/>Heatmap"] P58["5.4.8 Escalate High<br/>Risk to CAPA"] end subgraph DS_RISK["Data Stores"]
DR1[("risk_register")] DR2[("risk_assessments<br/>likelihood<br/>impact scores")]
DR3[("risk_matrix_config")] DR4[("mitigation_plans<br/>mitigation_actions")]
DR5[("risk_reviews<br/>residual_risk")] DR6[("risk_heatmap_cache")] end ER4 -->|"Risk
Identification<br/>(Hazard, Event, Context)"| P51 ER1 -->|"Risk Description,<br/>Category, Clinic Scope"|
P51 P51 --> DR1 P51 --> P52 ER1 -->|"Assessment Input<br/>(L × I Scores)"| P52 P52 --> DR2 P52 --> P53 P53
--> DR2 & DR3 P53 -->|"Risk Level<br/>(Low/Med/High/Extreme)"| ER1 P53 --> P54 ER1 & ER3 -->|"Mitigation
Strategy,<br/>Action Items, Timeline"| P54 P54 --> DR4 P54 --> P55 ER1 -->|"Implementation
Status,<br/>Evidence"| P55 P55 --> DR4 P55 --> P56 ER2 -->|"Periodic Review,<br/>Acceptance Decision"| P56
P56 --> DR5 P56 --> P57 P57 --> DR6 P57 -->|"Heatmap Data"| ER2 & ER3 P56 -->|"High/Extreme
Risk<br/>Unresolved"| P58 P58 -->|"CAPA Trigger"| CAPA_R["4.0 CAPA Module"] P58 --> DR1 ER2 -->|"Risk
Appetite<br/>Threshold Config"| DR3 classDef ext fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef
proc fill:#FFEBEE,stroke:#B71C1C,stroke-width:2px classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px
classDef cross fill:#FCE4EC,stroke:#C2185B,stroke-width:2px class ER1,ER2,ER3,ER4 ext class
P51,P52,P53,P54,P55,P56,P57,P58 proc class DR1,DR2,DR3,DR4,DR5,DR6 ds class CAPA_R cross --- %%
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ %% DFD
LEVEL 2 — ACCREDITATION MANAGEMENT %%
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
flowchart TB subgraph EXT_ACC["External Entities"] EX1["Accreditation Admin<br/>(Manajemen Pusat)"]
EX2["Clinic Assessor /<br/>PJ Mutu Klinik"] EX3["Unit Staff /<br/>Evidence Provider"]
EX4["Surveyors<br/>(Mock Survey Team)"] EX5["Kepala Klinik"] end subgraph P_ACC["6.0 Accreditation
Management"] direction TB P61["6.5.1 Configure Standards<br/>(FKTP / FKTL)"] P62["6.5.2 Manage
Hierarchy<br/>Bab → Standar → Kriteria<br/>→ Elemen Penilaian"] P63["6.5.3 Self
Assessment<br/>Execution"] P64["6.5.4 Score & Compliance<br/>Calculation"] P65["6.5.5 Gap Analysis<br/>&
Prioritization"] P66["6.5.6 Evidence Repository<br/>Management"] P67["6.5.7 Mock Survey<br/>Simulation"]
P68["6.5.8 Readiness Dashboard<br/>Aggregation"] end subgraph DS_ACC["Data Stores"]
DS1[("accreditation_standards<br/>standard_types: FKTP/FKTL")]
DS2[("standard_chapters<br/>standards<br/>criteria<br/>assessment_elements")]
DS3[("self_assessments<br/>assessment_responses")]
DS4[("gap_records<br/>gap_priority<br/>action_required")] DS5[("evidence_repository<br/>evidence_links")]
DS6[("mock_surveys<br/>survey_results")] DS7[("readiness_scores<br/>clinic_rankings")] end EX1
-->|"Add/Edit/Disable<br/>Standard Elements"| P61 P61 --> DS1 P61 --> P62 EX1 -->|"Hierarchy CRUD<br/>(No
Hardcoding)"| P62 P62 --> DS2 P62 --> P63 EX2 -->|"Compliance Response<br/>per Element (Ya/Tidak/NA)"| P63
EX3 -->|"Supporting Narrative"| P63 P63 --> DS3 P63 --> P64 P64 --> DS3 P64 -->|"Compliance %<br/>per
Standard/Bab"| EX2 P64 --> P65 EX2 -->|"Gap Confirmation,<br/>Root Cause, Priority"| P65 P65 --> DS4 P65
-->|"Gap Action Items"| CAPA_A["4.0 CAPA Module"] P63 --> P66 EX3 -->|"Upload Evidence<br/>(Doc, Photo,
Link)"| P66 P66 --> DS5 P66 -->|"Link to DMS<br/>Documents"| DMS_A["2.0 DMS Module"] P65 --> P67 EX4
-->|"Mock Survey Scoring,<br/>Observations"| P67 P67 --> DS6 P64 & P65 & P67 --> P68 P68 --> DS7 P68
-->|"Readiness Score,<br/>Clinic Ranking"| EX1 & EX5 EX5 -->|"Self Assessment<br/>Approval & Sign-off"|
P63 classDef ext fill:#E3F2FD,stroke:#1565C0,stroke-width:2px classDef proc
fill:#E0F7FA,stroke:#00695C,stroke-width:2px classDef ds fill:#FFFDE7,stroke:#F9A825,stroke-width:2px
classDef cross fill:#FFF3E0,stroke:#E65100,stroke-width:2px class EX1,EX2,EX3,EX4,EX5 ext class
P61,P62,P63,P64,P65,P66,P67,P68 proc class DS1,DS2,DS3,DS4,DS5,DS6,DS7 ds class CAPA_A,DMS_A cross
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ERD

[Diagram Mermaid — lihat file diagrams/*.mmd untuk versi visual]

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%% ═══════════════════════════════════════════════════════════════════════ %% SIMMAK PPKP — Enterprise Entity Relationship Diagram %% Sistem Informasi Manajemen Mutu, Audit Internal dan Akreditasi %% Organization: PPKP Provinsi DKI Jakarta | 17 Clinics | Single DB PostgreSQL 17 %% Version: 1.0.0 | Tables: 98 | Domains: 10 %% ═══════════════════════════════════════════════════════════════════════ %%{init: { 'theme': 'base', 'themeVariables': { 'primaryColor': '#E8F4FD', 'primaryTextColor': '#1A365D', 'primaryBorderColor': '#3182CE', 'lineColor': '#4A5568', 'secondaryColor': '#F0FFF4', 'tertiaryColor': '#FFFAF0', 'fontFamily': 'Segoe UI, Inter, sans-serif', 'fontSize': '13px' }, 'er': { 'useMaxWidth': true, 'layoutDirection': 'TB' } }}%% erDiagram %% ═══════════════════════════════════════════════════════════════════ %% DOMAIN 1: CORE PLATFORM (14 tables) — #E3F2FD %% Multi-scope: organization_id | clinic_id | unit_id on transactional data %% ═══════════════════════════════════════════════════════════════════ ORGANIZATIONS { uuid id PK varchar code UK "ORG code" varchar name "PPKP DKI Jakarta" varchar org_type "central|clinic" text address varchar phone varchar email boolean is_active timestamptz created_at timestamptz updated_at uuid created_by FK uuid updated_by FK } CLINICS { uuid id PK uuid organization_id FK varchar code UK "Klinik code" varchar name varchar clinic_type "utama|pratama" boolean is_main_clinic varchar accreditation_status boolean is_active timestamptz created_at timestamptz updated_at } DEPARTMENTS { uuid id PK uuid organization_id FK uuid parent_id FK "self-ref hierarchy" varchar code UK varchar name varchar dept_type boolean is_active } UNITS { uuid id PK uuid clinic_id FK uuid department_id FK varchar code UK varchar name varchar unit_type "poli|lab|farmasi|etc" boolean is_active } EMPLOYEES { uuid id PK uuid organization_id FK uuid clinic_id FK uuid unit_id FK varchar nip UK varchar full_name varchar email varchar phone varchar position_title date hire_date boolean is_active } REFERENCE_DATA { uuid id PK uuid organization_id FK varchar category "finding_severity|risk_level|etc" varchar code UK varchar label jsonb metadata int sort_order boolean is_active } REFERENCE_DATA_VALUES { uuid id PK uuid reference_data_id FK varchar code varchar label varchar color_hex int sort_order boolean is_default } FISCAL_YEARS { uuid id PK uuid organization_id FK int year UK date start_date date end_date boolean is_current varchar status "open|closed" } SYSTEM_SETTINGS { uuid id PK varchar group "app|mail|storage|ldap" varchar key UK text value varchar value_type "string|int|bool|json" boolean is_encrypted text description } ORGANIZATION_SETTINGS { uuid id PK uuid organization_id FK varchar key text value varchar value_type } CLINIC_SETTINGS { uuid id PK uuid clinic_id FK varchar key text value varchar value_type } AUDIT_TRAILS { uuid id PK uuid user_id FK varchar auditable_type uuid auditable_id varchar event "created|updated|deleted|restored" jsonb old_values jsonb new_values varchar ip_address timestamptz created_at } ACTIVITY_LOGS { uuid id PK uuid user_id FK uuid clinic_id FK varchar log_name varchar subject_type uuid subject_id text description jsonb properties varchar ip_address timestamptz created_at } ATTACHMENTS { uuid id PK uuid organization_id FK varchar attachable_type "polymorphic" uuid attachable_id varchar file_name varchar storage_disk "supabase|local" varchar storage_path varchar mime_type bigint file_size uuid uploaded_by FK timestamptz created_at } %% ═══════════════════════════════════════════════════════════════════ %% DOMAIN 2: AUTH & RBAC (11 tables) — #FFF3E0 %% Hybrid: Local Login + LDAP/AD Pemprov DKI Jakarta %% ═══════════════════════════════════════════════════════════════════ USERS { uuid id PK uuid organization_id FK uuid employee_id FK varchar email UK varchar password "nullable for LDAP-only" varchar ldap_guid UK varchar auth_provider "local|ldap|hybrid" boolean is_active boolean force_password_change timestamptz email_verified_at timestamptz last_login_at timestamptz created_at timestamptz updated_at } ROLES { uuid id PK uuid organization_id FK varchar name UK varchar slug UK text description varchar scope "global|clinic|unit" boolean is_system boolean is_active } PERMISSIONS { uuid id PK varchar module "dms|audit|capa|risk|accreditation|quality" varchar action "view|create|update|delete|approve|publish" varchar slug UK varchar name text description } ROLE_PERMISSIONS { uuid role_id PK,FK uuid permission_id PK,FK timestamptz granted_at } USER_ROLES { uuid id PK uuid user_id FK uuid role_id FK uuid clinic_id FK "nullable = global scope" uuid unit_id FK timestamptz assigned_at uuid assigned_by FK } USER_CLINIC_SCOPES { uuid id PK uuid user_id FK uuid clinic_id FK boolean is_default boolean can_view boolean can_edit } LDAP_CONFIGURATIONS { uuid id PK uuid organization_id FK varchar host int port varchar base_dn varchar bind_dn text bind_password_enc varchar user_filter varchar group_filter boolean use_tls boolean is_active timestamptz last_sync_at } LDAP_SYNC_LOGS { uuid id PK uuid ldap_configuration_id FK varchar sync_type "full|incremental" varchar status "running|success|failed" int users_created int users_updated int users_deactivated text error_message timestamptz started_at timestamptz finished_at } LOGIN_ATTEMPTS { uuid id PK varchar email varchar ip_address varchar user_agent boolean successful varchar failure_reason timestamptz attempted_at } SESSIONS { uuid id PK uuid user_id FK varchar ip_address text user_agent text payload int last_activity } PERSONAL_ACCESS_TOKENS { uuid id PK uuid user_id FK varchar name varchar token UK text abilities timestamptz last_used_at timestamptz expires_at } %% ═══════════════════════════════════════════════════════════════════ %% DOMAIN 3: DOCUMENT MANAGEMENT SYSTEM (12 tables) — #E8F5E9 %% Kebijakan | SOP | Pedoman | Formulir | IK | Version | QR | Expiry %% ═══════════════════════════════════════════════════════════════════ DOCUMENT_CATEGORIES { uuid id PK uuid organization_id FK uuid parent_id FK varchar code UK varchar name varchar doc_class
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"kebijakan|sop|pedoman|formulir|ik" int retention_days boolean is_active } DOCUMENT_TYPES { uuid id PK
uuid organization_id FK uuid category_id FK varchar code UK varchar name uuid workflow_definition_id FK
int review_cycle_days boolean requires_acknowledgment boolean is_active } DOCUMENTS { uuid id PK uuid
organization_id FK uuid clinic_id FK "nullable = org-wide" uuid document_type_id FK uuid
workflow_instance_id FK varchar document_number UK varchar title varchar status
"draft|review|approved|published|expired|archived" date effective_date date expiry_date uuid owner_id FK
uuid current_version_id FK timestamptz created_at timestamptz updated_at } DOCUMENT_VERSIONS { uuid id PK
uuid document_id FK varchar version_label "v1.0|v1.1" int version_number text change_summary text
content_html varchar file_path "Supabase Storage" varchar checksum_sha256 uuid created_by FK timestamptz
created_at } DOCUMENT_REVIEWS { uuid id PK uuid document_version_id FK uuid reviewer_id FK varchar status
"pending|approved|rejected|revision" text comments timestamptz reviewed_at } DOCUMENT_APPROVALS { uuid id
PK uuid document_version_id FK uuid approver_id FK varchar approval_level "reviewer|pj_mutu|kepala_klinik"
varchar status "pending|approved|rejected" text comments timestamptz approved_at } DOCUMENT_PUBLICATIONS {
uuid id PK uuid document_id FK uuid document_version_id FK uuid published_by FK timestamptz published_at
date effective_from varchar publication_note } DOCUMENT_DISTRIBUTIONS { uuid id PK uuid document_id FK
uuid target_clinic_id FK uuid distributed_by FK timestamptz distributed_at varchar distribution_method
"system|email|print" } DOCUMENT_ACKNOWLEDGMENTS { uuid id PK uuid document_id FK uuid user_id FK uuid
clinic_id FK timestamptz acknowledged_at varchar ip_address } DOCUMENT_QR_CODES { uuid id PK uuid
document_id FK varchar qr_token UK varchar verification_url varchar qr_image_path boolean is_active
timestamptz generated_at } DOCUMENT_EXPIRY_ALERTS { uuid id PK uuid document_id FK int days_before_expiry
varchar alert_status "pending|sent|acknowledged" timestamptz alert_sent_at uuid notified_user_id FK }
DOCUMENT_ACCESS_LOGS { uuid id PK uuid document_id FK uuid user_id FK varchar action "view|download|print"
varchar ip_address timestamptz accessed_at } %%
============================================================================ %% DOMAIN 4: INTERNAL AUDIT (11
tables) — #F3E5F5 %% Program | Schedule | Checklist | Finding | Report | Evidence %%
============================================================================ AUDIT_PROGRAMS { uuid id PK uuid
organization_id FK uuid fiscal_year_id FK varchar program_code UK varchar name text objective varchar
status "draft|active|completed|archived" uuid created_by FK timestamptz created_at } AUDIT_PLANS { uuid id
PK uuid audit_program_id FK uuid clinic_id FK uuid unit_id FK varchar plan_code UK text scope text
criteria_reference varchar status "planned|in_progress|completed" uuid lead_auditor_id FK }
AUDIT_SCHEDULES { uuid id PK uuid audit_plan_id FK date scheduled_start date scheduled_end date
actual_start date actual_end varchar status "scheduled|ongoing|completed|cancelled" text location }
AUDIT_TEAMS { uuid id PK uuid audit_schedule_id FK uuid lead_auditor_id FK text team_notes }
AUDIT_TEAM_MEMBERS { uuid id PK uuid audit_team_id FK uuid user_id FK varchar member_role
"auditor|observer|technical" } AUDIT_CHECKLIST_ITEMS { uuid id PK uuid audit_plan_id FK varchar
standard_reference text question varchar response_type "yes_no|score|text" decimal weight int sort_order }
AUDIT_CHECKLIST_RESPONSES { uuid id PK uuid checklist_item_id FK uuid audit_schedule_id FK varchar
response_value decimal score text notes uuid responded_by FK timestamptz responded_at } AUDIT_FINDINGS {
uuid id PK uuid audit_schedule_id FK uuid clinic_id FK varchar finding_code UK varchar severity
"minor|major|critical|observation" text description text requirement_reference varchar status
"open|in_capa|closed|verified" uuid assigned_to FK timestamptz identified_at } AUDIT_REPORTS { uuid id PK
uuid audit_schedule_id FK varchar report_number UK text executive_summary text conclusion varchar status
"draft|review|approved|published" uuid approved_by FK timestamptz approved_at } AUDIT_EVIDENCE { uuid id
PK uuid audit_finding_id FK uuid attachment_id FK text description varchar evidence_type
"photo|document|interview" uuid collected_by FK timestamptz collected_at } AUDIT_FOLLOW_UPS { uuid id PK
uuid audit_finding_id FK date due_date varchar status "pending|overdue|completed" uuid assigned_to FK text
follow_up_notes timestamptz completed_at } %%
============================================================================ %% DOMAIN 5: CAPA (9 tables) —
#FFEBEE %% RCA | 5 Why | Fishbone | Corrective | Preventive | Verification %%
============================================================================ CAPA_CASES { uuid id PK uuid
organization_id FK uuid clinic_id FK varchar case_number UK varchar source_type
"audit_finding|risk|complaint|incident" uuid source_id "polymorphic" varchar priority
"low|medium|high|critical" text problem_statement varchar status
"open|analysis|action|verification|closed" uuid owner_id FK date target_close_date timestamptz created_at
} CAPA_ROOT_CAUSE_ANALYSES { uuid id PK uuid capa_case_id FK varchar method "5why|fishbone|combined" text
summary uuid analyzed_by FK timestamptz analyzed_at } CAPA_FIVE_WHYS { uuid id PK uuid rca_id FK int
why_level "1-5" text question text answer } CAPA_FISHBONE_DIAGRAMS { uuid id PK uuid rca_id FK varchar
category "man|machine|method|material|environment|measurement" text cause_description boolean
is_root_cause } CAPA_CORRECTIVE_ACTIONS { uuid id PK uuid capa_case_id FK text action_description uuid
responsible_id FK date planned_start date due_date varchar status "planned|in_progress|completed|overdue"
timestamptz completed_at } CAPA_PREVENTIVE_ACTIONS { uuid id PK uuid capa_case_id FK text
action_description uuid responsible_id FK date due_date varchar status "planned|in_progress|completed"
timestamptz completed_at } CAPA_VERIFICATIONS { uuid id PK uuid capa_case_id FK uuid verified_by FK date
verification_date varchar result "effective|partially|not_effective" text verification_notes uuid
evidence_attachment_id FK } CAPA_EFFECTIVENESS_REVIEWS { uuid id PK uuid capa_case_id FK uuid reviewed_by
```

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FK date review_date boolean is_effective text review_notes date next_review_date } CAPA_STATUS_HISTORIES {
uuid id PK uuid capa_case_id FK varchar from_status varchar to_status uuid changed_by FK text
change_reason timestamptz changed_at } %%
========================================================================== %% DOMAIN 6: RISK MANAGEMENT (8
tables) — #FFF8E1 %% Register | Assessment | Matrix | Mitigation | Monitoring %%
========================================================================== RISK_CATEGORIES { uuid id PK
uuid organization_id FK varchar code UK varchar name varchar color_hex text description boolean is_active
} RISK_REGISTERS { uuid id PK uuid organization_id FK uuid clinic_id FK uuid fiscal_year_id FK varchar
register_code UK varchar name varchar status "draft|active|review|archived" uuid owner_id FK } RISK_ITEMS
{ uuid id PK uuid risk_register_id FK uuid risk_category_id FK varchar risk_code UK text risk_description
text risk_source int inherent_likelihood int inherent_impact int inherent_score int residual_likelihood
int residual_impact int residual_score varchar risk_level "low|medium|high|extreme" varchar status
"identified|assessed|mitigated|accepted|closed" } RISK_ASSESSMENTS { uuid id PK uuid risk_item_id FK uuid
assessed_by FK date assessment_date int likelihood int impact int risk_score text assessment_notes varchar
assessment_type "inherent|residual" } RISK_MATRICES { uuid id PK uuid organization_id FK varchar name
jsonb likelihood_levels jsonb impact_levels jsonb risk_level_mapping boolean is_default }
RISK_MITIGATION_PLANS { uuid id PK uuid risk_item_id FK varchar strategy "avoid|reduce|transfer|accept"
text plan_description uuid responsible_id FK date target_date varchar status
"planned|in_progress|completed" decimal estimated_cost } RISK_MONITORING_LOGS { uuid id PK uuid
risk_item_id FK uuid monitored_by FK date monitoring_date int current_likelihood int current_impact text
monitoring_notes varchar trend "increasing|stable|decreasing" } RISK_REVIEWS { uuid id PK uuid
risk_register_id FK date review_date uuid reviewed_by FK text conclusion varchar review_outcome
"accept|revise|escalate" date next_review_date } %%
========================================================================== %% DOMAIN 7: ACCREDITATION
MANAGEMENT (12 tables) — #E0F7FA %% FKTP & FKTL | Dynamic Standards | Self Assessment | Gap | Mock Survey
%% ========================================================================== ACCREDITATION_TYPES { uuid id
PK varchar code UK "FKTP|FKTL" varchar name text description varchar issuing_body "KARS|etc" boolean
is_active } ACCREDITATION_CHAPTERS { uuid id PK uuid accreditation_type_id FK varchar chapter_number
varchar title text description int sort_order boolean is_active } ACCREDITATION_STANDARDS { uuid id PK
uuid chapter_id FK varchar standard_code UK varchar title text description int sort_order boolean
is_active } ACCREDITATION_CRITERIA { uuid id PK uuid standard_id FK varchar criteria_code UK text
description decimal weight int sort_order boolean is_active } ACCREDITATION_ELEMENTS { uuid id PK uuid
criteria_id FK varchar element_code UK text description text evidence_required int sort_order boolean
is_mandatory boolean is_active } ACCREDITATION_PROGRAMS { uuid id PK uuid organization_id FK uuid
accreditation_type_id FK varchar program_name varchar cycle "2024-2027" date start_date date
target_survey_date varchar status "planning|self_assessment|mock|survey|accredited" }
ACCREDITATION_SELF_ASSESSMENTS { uuid id PK uuid program_id FK uuid clinic_id FK date assessment_date uuid
assessor_id FK decimal overall_score varchar status "draft|submitted|reviewed|approved" text summary }
ACCREDITATION_SELF_ASSESSMENT_ITEMS { uuid id PK uuid self_assessment_id FK uuid element_id FK decimal
score "0-100" varchar compliance "compliant|partial|non_compliant|na" text notes text improvement_plan }
ACCREDITATION_GAP_ANALYSES { uuid id PK uuid self_assessment_id FK uuid analyzed_by FK date analysis_date
text executive_summary varchar status "draft|final" } ACCREDITATION_GAP_ITEMS { uuid id PK uuid
gap_analysis_id FK uuid element_id FK text gap_description varchar priority "low|medium|high|critical"
text recommended_action uuid assigned_to FK date target_date } ACCREDITATION_MOCK_SURVEYS { uuid id PK
uuid program_id FK uuid clinic_id FK date survey_date uuid lead_surveyor_id FK decimal readiness_score
text findings_summary varchar status "scheduled|completed|reported" } ACCREDITATION_EVIDENCE { uuid id PK
uuid clinic_id FK uuid element_id FK uuid attachment_id FK varchar title text description date valid_from
date valid_until uuid uploaded_by FK timestamptz uploaded_at } %%
========================================================================== %% DOMAIN 8: QUALITY MANAGEMENT
(7 tables) — #F1F8E9 %% Program Mutu | Indicators | Monitoring | Evaluation %%
========================================================================== QUALITY_PROGRAMS { uuid id PK
uuid organization_id FK uuid fiscal_year_id FK varchar program_code UK varchar name text objective varchar
status "draft|active|evaluation|closed" uuid program_owner_id FK } QUALITY_INDICATORS { uuid id PK uuid
program_id FK varchar indicator_code UK varchar name text definition varchar unit "percent|count|ratio"
varchar direction "higher_better|lower_better" decimal baseline_value varchar data_source }
QUALITY_INDICATOR_TARGETS { uuid id PK uuid indicator_id FK uuid clinic_id FK varchar period
"monthly|quarterly|annual" decimal target_value decimal threshold_warning decimal threshold_critical }
QUALITY_MEASUREMENTS { uuid id PK uuid indicator_id FK uuid clinic_id FK date measurement_date decimal
measured_value varchar period_label "2026-Q1" uuid recorded_by FK text notes } QUALITY_MONITORING_SESSIONS
{ uuid id PK uuid program_id FK uuid clinic_id FK date session_date uuid facilitator_id FK text findings
varchar status "scheduled|completed" } QUALITY_EVALUATIONS { uuid id PK uuid program_id FK uuid clinic_id
FK date evaluation_date decimal overall_score text strengths text weaknesses uuid evaluator_id FK }
QUALITY_REPORTS { uuid id PK uuid program_id FK varchar report_period uuid generated_by FK varchar
file_path timestamptz generated_at } %%
========================================================================== %% DOMAIN 9: WORKFLOW ENGINE (8
```

```
tables) — #EDE7F6 %% Dynamic configurable workflows — no hardcoded approval chains %%
========================================================================== WORKFLOW_DEFINITIONS { uuid id
PK uuid organization_id FK varchar module "dms|audit|capa|accreditation" varchar name varchar code UK int
version boolean is_active jsonb configuration uuid created_by FK } WORKFLOW_STEPS { uuid id PK uuid
workflow_definition_id FK int step_order varchar step_code varchar name varchar step_type
"review|approve|publish|notify" uuid required_role_id FK int sla_hours boolean is_parallel }
WORKFLOW_TRANSITIONS { uuid id PK uuid workflow_definition_id FK uuid from_step_id FK uuid to_step_id FK
varchar action "approve|reject|revise|skip" jsonb conditions } WORKFLOW_INSTANCES { uuid id PK uuid
workflow_definition_id FK varchar subject_type "Document|CapaCase|etc" uuid subject_id uuid
current_step_id FK varchar status "running|completed|cancelled|escalated" uuid initiated_by FK timestamptz
started_at timestamptz completed_at } WORKFLOW_INSTANCE_STEPS { uuid id PK uuid workflow_instance_id FK
uuid workflow_step_id FK varchar status "pending|in_progress|completed|skipped" uuid assignee_id FK
timestamptz started_at timestamptz completed_at int sla_breached_hours } WORKFLOW_ASSIGNMENTS { uuid id PK
uuid instance_step_id FK uuid assignee_id FK uuid assigned_by FK timestamptz assigned_at timestamptz
due_at } WORKFLOW_APPROVALS { uuid id PK uuid instance_step_id FK uuid approver_id FK varchar decision
"approved|rejected|revision_requested" text comments timestamptz decided_at } WORKFLOW_HISTORIES { uuid id
PK uuid workflow_instance_id FK uuid from_step_id FK uuid to_step_id FK varchar action uuid action_by FK
jsonb metadata timestamptz action_at } %%
========================================================================== %% DOMAIN 10: NOTIFICATION (6
tables) — #ECEFF1 %% Multi-channel: in-app | email | (future: SMS/WhatsApp) %%
========================================================================== NOTIFICATION_CHANNELS { uuid id
PK varchar code UK "in_app|email|sms" varchar name jsonb configuration boolean is_enabled }
NOTIFICATION_TEMPLATES { uuid id PK uuid channel_id FK varchar event_type UK
"document_expiry|capa_due|audit_scheduled" varchar subject_template text body_template jsonb
variables_schema boolean is_active } NOTIFICATIONS { uuid id PK uuid organization_id FK uuid clinic_id FK
varchar type varchar title text body jsonb data varchar priority "low|normal|high|urgent" timestamptz
sent_at uuid triggered_by FK } NOTIFICATION_RECIPIENTS { uuid id PK uuid notification_id FK uuid user_id
FK varchar delivery_status "pending|sent|delivered|failed" timestamptz read_at timestamptz delivered_at }
NOTIFICATION_PREFERENCES { uuid id PK uuid user_id FK varchar event_type uuid channel_id FK boolean
is_enabled varchar digest_frequency "instant|daily|weekly" } NOTIFICATION_LOGS { uuid id PK uuid
notification_id FK uuid channel_id FK varchar status "queued|sent|failed|bounced" text error_message jsonb
provider_response timestamptz attempted_at } %%
========================================================================== %% RELATIONSHIPS — CORE PLATFORM
%% ========================================================================== ORGANIZATIONS ||--o{ CLINICS
: "operates" ORGANIZATIONS ||--o{ DEPARTMENTS : "has" ORGANIZATIONS ||--o{ EMPLOYEES : "employs"
ORGANIZATIONS ||--o{ REFERENCE_DATA : "defines" ORGANIZATIONS ||--o{ FISCAL_YEARS : "plans" ORGANIZATIONS
||--o{ ORGANIZATION_SETTINGS : "configures" CLINICS ||--o{ UNITS : "contains" CLINICS ||--o{
CLINIC_SETTINGS : "configures" CLINICS ||--o{ EMPLOYEES : "assigns" DEPARTMENTS ||--o{ UNITS : "groups"
DEPARTMENTS ||--o{ DEPARTMENTS : "parent_of" REFERENCE_DATA ||--o{ REFERENCE_DATA_VALUES : "has_values"
EMPLOYEES ||--o| USERS : "linked_account" %% RELATIONSHIPS — AUTH & RBAC ORGANIZATIONS ||--o{ USERS :
"has_users" ORGANIZATIONS ||--o{ ROLES : "defines_roles" ORGANIZATIONS ||--o{ LDAP_CONFIGURATIONS :
"integrates" USERS ||--o{ USER_ROLES : "assigned" USERS ||--o{ USER_CLINIC_SCOPES : "scoped_to" USERS
||--o{ SESSIONS : "maintains" USERS ||--o{ LOGIN_ATTEMPTS : "attempts" USERS ||--o{ PERSONAL_ACCESS_TOKENS
: "api_access" ROLES ||--o{ ROLE_PERMISSIONS : "grants" PERMISSIONS ||--o{ ROLE_PERMISSIONS :
"granted_via" ROLES ||--o{ USER_ROLES : "applied" CLINICS ||--o{ USER_CLINIC_SCOPES : "accessible" CLINICS
||--o{ USER_ROLES : "scoped" LDAP_CONFIGURATIONS ||--o{ LDAP_SYNC_LOGS : "logs" USERS ||--o{ AUDIT_TRAILS
: "performs" USERS ||--o{ ACTIVITY_LOGS : "generates" %% RELATIONSHIPS — DMS ORGANIZATIONS ||--o{
DOCUMENT_CATEGORIES : "classifies" ORGANIZATIONS ||--o{ DOCUMENT_TYPES : "defines" DOCUMENT_CATEGORIES
||--o{ DOCUMENT_TYPES : "categorizes" DOCUMENT_TYPES ||--o{ DOCUMENTS : "templates" CLINICS ||--o{
DOCUMENTS : "owns" DOCUMENTS ||--o{ DOCUMENT_VERSIONS : "versions" DOCUMENTS ||--o| DOCUMENT_VERSIONS :
"current_version" DOCUMENT_VERSIONS ||--o{ DOCUMENT_REVIEWS : "reviewed" DOCUMENT_VERSIONS ||--o{
DOCUMENT_APPROVALS : "approved" DOCUMENTS ||--o{ DOCUMENT_PUBLICATIONS : "published" DOCUMENTS ||--o{
DOCUMENT_DISTRIBUTIONS : "distributed" DOCUMENTS ||--o{ DOCUMENT_ACKNOWLEDGMENTS : "acknowledged"
DOCUMENTS ||--o| DOCUMENT_QR_CODES : "verified" DOCUMENTS ||--o{ DOCUMENT_EXPIRY_ALERTS : "monitors"
DOCUMENTS ||--o{ DOCUMENT_ACCESS_LOGS : "tracked" USERS ||--o{ DOCUMENT_REVIEWS : "reviews" USERS ||--o{
DOCUMENT_APPROVALS : "approves" WORKFLOW_INSTANCES ||--o| DOCUMENTS : "governs" %% RELATIONSHIPS —
INTERNAL AUDIT ORGANIZATIONS ||--o{ AUDIT_PROGRAMS : "plans" FISCAL_YEARS ||--o{ AUDIT_PROGRAMS : "within"
AUDIT_PROGRAMS ||--o{ AUDIT_PLANS : "contains" CLINICS ||--o{ AUDIT_PLANS : "audited" AUDIT_PLANS ||--o{
AUDIT_SCHEDULES : "scheduled" AUDIT_SCHEDULES ||--o| AUDIT_TEAMS : "staffed" AUDIT_TEAMS ||--o{
AUDIT_TEAM_MEMBERS : "includes" AUDIT_PLANS ||--o{ AUDIT_CHECKLIST_ITEMS : "checklists"
AUDIT_CHECKLIST_ITEMS ||--o{ AUDIT_CHECKLIST_RESPONSES : "answered" AUDIT_SCHEDULES ||--o{
AUDIT_CHECKLIST_RESPONSES : "during" AUDIT_SCHEDULES ||--o{ AUDIT_FINDINGS : "identifies" AUDIT_SCHEDULES
||--o| AUDIT_REPORTS : "reports" AUDIT_FINDINGS ||--o{ AUDIT_EVIDENCE : "supports" AUDIT_FINDINGS ||--o{
AUDIT_FOLLOW_UPS : "tracks" ATTACHMENTS ||--o{ AUDIT_EVIDENCE : "stores" %% RELATIONSHIPS — CAPA
ORGANIZATIONS ||--o{ CAPA_CASES : "manages" CLINICS ||--o{ CAPA_CASES : "owns" AUDIT_FINDINGS ||--o{
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CAPA_CASES : "triggers" CAPA_CASES ||--o| CAPA_ROOT_CAUSE_ANALYSES : "analyzed" CAPA_ROOT_CAUSE_ANALYSES
||--o{ CAPA_FIVE_WHYS : "5why_chain" CAPA_ROOT_CAUSE_ANALYSES ||--o{ CAPA_FISHBONE_DIAGRAMS : "fishbone"
CAPA_CASES ||--o{ CAPA_CORRECTIVE_ACTIONS : "corrects" CAPA_CASES ||--o{ CAPA_PREVENTIVE_ACTIONS :
"prevents" CAPA_CASES ||--o{ CAPA_VERIFICATIONS : "verifies" CAPA_CASES ||--o{ CAPA_EFFECTIVENESS_REVIEWS
: "reviews" CAPA_CASES ||--o{ CAPA_STATUS_HISTORIES : "audited" %% RELATIONSHIPS — RISK ORGANIZATIONS
||--o{ RISK_CATEGORIES : "classifies" ORGANIZATIONS ||--o{ RISK_REGISTERS : "maintains" ORGANIZATIONS
||--o{ RISK_MATRICES : "configures" CLINICS ||--o{ RISK_REGISTERS : "scoped" FISCAL_YEARS ||--o{
RISK_REGISTERS : "annual" RISK_REGISTERS ||--o{ RISK_ITEMS : "contains" RISK_CATEGORIES ||--o{ RISK_ITEMS
: "categorizes" RISK_ITEMS ||--o{ RISK_ASSESSMENTS : "assessed" RISK_ITEMS ||--o{ RISK_MITIGATION_PLANS :
"mitigated" RISK_ITEMS ||--o{ RISK_MONITORING_LOGS : "monitored" RISK_REGISTERS ||--o{ RISK_REVIEWS :
"reviewed" %% RELATIONSHIPS — ACCREDITATION ACCREDITATION_TYPES ||--o{ ACCREDITATION_CHAPTERS :
"structured" ACCREDITATION_CHAPTERS ||--o{ ACCREDITATION_STANDARDS : "contains" ACCREDITATION_STANDARDS
||--o{ ACCREDITATION_CRITERIA : "defines" ACCREDITATION_CRITERIA ||--o{ ACCREDITATION_ELEMENTS :
"breaks_down" ORGANIZATIONS ||--o{ ACCREDITATION_PROGRAMS : "runs" ACCREDITATION_TYPES ||--o{
ACCREDITATION_PROGRAMS : "targets" ACCREDITATION_PROGRAMS ||--o{ ACCREDITATION_SELF_ASSESSMENTS :
"assesses" CLINICS ||--o{ ACCREDITATION_SELF_ASSESSMENTS : "evaluated" ACCREDITATION_SELF_ASSESSMENTS
||--o{ ACCREDITATION_SELF_ASSESSMENT_ITEMS : "scores" ACCREDITATION_ELEMENTS ||--o{
ACCREDITATION_SELF_ASSESSMENT_ITEMS : "rated" ACCREDITATION_SELF_ASSESSMENTS ||--o|
ACCREDITATION_GAP_ANALYSES : "gaps" ACCREDITATION_GAP_ANALYSES ||--o{ ACCREDITATION_GAP_ITEMS : "items"
ACCREDITATION_PROGRAMS ||--o{ ACCREDITATION_MOCK_SURVEYS : "simulates" CLINICS ||--o{
ACCREDITATION_EVIDENCE : "provides" ACCREDITATION_ELEMENTS ||--o{ ACCREDITATION_EVIDENCE : "linked"
ATTACHMENTS ||--o{ ACCREDITATION_EVIDENCE : "stores" %% RELATIONSHIPS — QUALITY ORGANIZATIONS ||--o{
QUALITY_PROGRAMS : "runs" FISCAL_YEARS ||--o{ QUALITY_PROGRAMS : "within" QUALITY_PROGRAMS ||--o{
QUALITY_INDICATORS : "measures" QUALITY_INDICATORS ||--o{ QUALITY_INDICATOR_TARGETS : "targets"
QUALITY_INDICATORS ||--o{ QUALITY_MEASUREMENTS : "records" CLINICS ||--o{ QUALITY_INDICATOR_TARGETS :
"clinic_target" CLINICS ||--o{ QUALITY_MEASUREMENTS : "reports" QUALITY_PROGRAMS ||--o{
QUALITY_MONITORING_SESSIONS : "monitors" QUALITY_PROGRAMS ||--o{ QUALITY_EVALUATIONS : "evaluates"
QUALITY_PROGRAMS ||--o{ QUALITY_REPORTS : "reports" %% RELATIONSHIPS — WORKFLOW ORGANIZATIONS ||--o{
WORKFLOW_DEFINITIONS : "configures" WORKFLOW_DEFINITIONS ||--o{ WORKFLOW_STEPS : "steps"
WORKFLOW_DEFINITIONS ||--o{ WORKFLOW_TRANSITIONS : "transitions" WORKFLOW_DEFINITIONS ||--o{
WORKFLOW_INSTANCES : "executes" WORKFLOW_INSTANCES ||--o{ WORKFLOW_INSTANCE_STEPS : "progress"
WORKFLOW_STEPS ||--o{ WORKFLOW_INSTANCE_STEPS : "instantiated" WORKFLOW_INSTANCE_STEPS ||--o{
WORKFLOW_ASSIGNMENTS : "assigned" WORKFLOW_INSTANCE_STEPS ||--o{ WORKFLOW_APPROVALS : "decided"
WORKFLOW_INSTANCES ||--o{ WORKFLOW_HISTORIES : "logged" DOCUMENT_TYPES ||--o| WORKFLOW_DEFINITIONS :
"uses" ROLES ||--o{ WORKFLOW_STEPS : "required_role" %% RELATIONSHIPS — NOTIFICATION NOTIFICATION_CHANNELS
||--o{ NOTIFICATION_TEMPLATES : "templates" ORGANIZATIONS ||--o{ NOTIFICATIONS : "broadcasts"
NOTIFICATIONS ||--o{ NOTIFICATION_RECIPIENTS : "delivers" NOTIFICATIONS ||--o{ NOTIFICATION_LOGS : "logs"
USERS ||--o{ NOTIFICATION_RECIPIENTS : "receives" USERS ||--o{ NOTIFICATION_PREFERENCES : "prefers"
NOTIFICATION_CHANNELS ||--o{ NOTIFICATION_PREFERENCES : "via" NOTIFICATION_CHANNELS ||--o{
NOTIFICATION_LOGS : "channel"
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